Mango-MT: Advancing Audiovisual Translation with a 9B Multilingual Model and Expert-Annotated Benchmark

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June 2026

Abstract

Global distribution of long-form videos demands reliable multilingual subtitle translation, yet generic translation models struggle with subtitles' fragmented text, strict timeline rules and plot-dependent context, yielding unstable outputs in mass production. To solve these industrial pain points, we present Mango-MT with 9B parameter size, an 11-language translation model tailored for audiovisual subtitles, alongside a dedicated evaluation benchmark called Mango-SubBench. The base model is pre-trained on diverse high-quality datasets covering 11 languages to fully unlock the potential of multilingual data. We then perform instruction tuning on the pre-trained model, followed by further optimization via reinforcement learning (RL), which greatly boosts the model's generalization across various language pairs. Tests across 11 languages prove Mango-MT surpasses mainstream models like GPT, Gemini and DeepSeek, with robust retention of subtitle timelines, unified semantics and scalable production capacity. We fully release the optimal practices derived from our optimization pipeline and open-source the model weights to facilitate research and industrial applications in multilingual translation. Code is available at `https://github.com/MGTVAI/Mango-MT`.

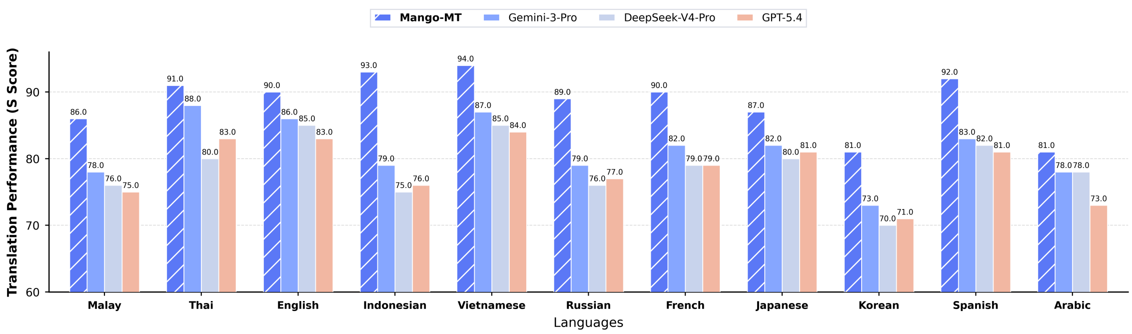

Figure 1: Overall Score of 11 Languages on the Mango-SubBench

1 Introduction

Machine translation has long been a practically valuable research target with extensive real-world demand, while remaining a challenging research problem over the past several decades [Brown et al. [1993];Wu et al. [2016];Vaswani et al. [2017]]. In recent years, the advancement of large language models (LLMs) has thoroughly revolutionized the learning paradigm of machine translation and steered the field toward large-scale learning pipelines [Garcia et al. [2023];Alves et al. [2024]]. Continuous iterations of LLMs keep raising the ceiling of translation quality;

state-of-the-art models such as Gemini-3-Pro [DeepMind [2025]] even outperform professional human translators on certain language translation directions [Kocmi et al. [2024]]

Nevertheless, these underlying challenges remain far from fully resolved, especially in multilingual audiovisual subtitle translation. Core pain points include missing plot context, fragmented subtitle utterances, rigid timeline formatting constraints, and heavy reliance on global story context for coherent translation outputs. These issues collectively lead to unstable translation quality in mass production pipelines. Furthermore, a persistent performance gap has long separated closed-source commercial models from open-source alternatives [Kocmi et al. [2024]]. Open-source systems are constrained by limited model parameter scales and insufficient translation training data, a limitation that is particularly pronounced in multilingual subtitle translation due to the scarcity of high-quality audiovisual parallel corpora. To date, the field lacks a mature, comprehensive technical framework for building state-of-the-art LLM-powered audiovisual translation systems, hindering researchers and developers from deploying and iterating high-performance translation pipelines. Equally absent is a dedicated, standardized benchmark tailored exclusively for evaluating audiovisual subtitle translation quality.

The training pipeline consists of three stages. First, we adopt the open-source Qwen3.5-9B as the base model and conduct domain-adaptive continued pre-training tailored for multilingual audiovisual subtitle translation. This stage integrates monolingual text, general multilingual corpora, and high-quality parallel data with a total volume of approximately 200B tokens, to strengthen the model's modeling capacity for film dialogues, colloquial expressions, cross-lingual mapping, and representations of low-resource languages. Second, we perform supervised fine-tuning on 7 million pieces of high-quality human-refined parallel corpora, enabling the model to learn subtitle translation instructions, contextual inputs, and structured JSON output formats. Third, we adopt a hybrid reinforcement learning alignment strategy that combines human preference feedback with automatic rewards derived from COMET and BLEU metrics. This further improves the model's semantic accuracy, contextual consistency, compliance with subtitle formatting rules, and its ability to generate expressive and vivid film dialogues. Moreover, mainstream general translation evaluation datasets mainly focus on sentence-level semantic alignment of generic text, while ignoring the unique scenario characteristics of audiovisual subtitles: fragmented utterances, strict layout and timeline constraints, and heavy reliance on plot context. Consequently, these benchmarks cannot faithfully reflect the practical translation performance of models under industrial mass-production pipelines. To address this limitation, we construct Mango-SubBench, a fine-grained multi-dimensional evaluation benchmark dedicated to audiovisual subtitle translation.

We conduct comprehensive comparative experiments on Mango-SubBench as well as mainstream translation benchmarks, with comparison baselines including various closed-source commercial translation models: DeepSeek-v4-Pro[DeepSeek-AI [2026]], Gemini-3 -Pro[DeepMind [2025]], and GPT-5.4[OpenAI [2026]]. Experimental results demonstrate that Mango-MT comprehensively outperforms all closed-source commercial counterparts across all 11 languages on Mango-SubBench, our audiovisual subtitle translation benchmark. Meanwhile, on the widely adopted Flores+ [NLLB Team et al. [2024]] benchmark, Mango-MT achieves translation performance matching or exceeding that of the closed-source commercial models. Overall, our work establishes an improved state-of-the-art baseline for machine translation research built upon open-source 9B-parameter models via the following contributions:

- Contribution 1: Mango-MT with 9B parameter size, the industry's first professional multilingual machine translation model exclusively designed for audiovisual content.

- Contribution 2: We propose Mango-SubBench, a novel benchmark for audiovisual subtitle translation, which evaluates translation quality from fine-grained multi-dimensional perspectives.

- Contribution 3: A complete training recipe for large language models specialized in multilingual audiovisual subtitle translation.

2 Data

2.1 Data Collection

We adopt a hybrid data construction strategy combining open-source parallel corpora with proprietary human-polished translation datasets to build a high-quality multilingual training corpus optimized for audiovisual subtitle translation scenarios.

2.1.1 Open-Source Parallel Corpora

For open-source data, we select multiple subsets from OPUS (Open Parallel Corpus)[Tiedemann [2012]] as our base data sources, with a primary focus on OpenSubtitles[Lison and Tiedemann [2016]], CCAligned[El-Kishky et al. [2020]], and WikiMatrix[Schwenk et al. [2021]] . As one of the most widely used subtitle-level parallel corpora within OPUS, OpenSubtitles consists of multilingual text extracted from film and television subtitles. By contrast, CCAligned and WikiMatrix are parallel datasets automatically mined from large-scale web corpora and Wikipedia text. Collectively, these datasets furnish abundant foundational translation examples for model training. We curate 200B tokens of open-source parallel data for pre-training.

2.1.2 In-House Mango Audiovisual Dataset

Beyond open-source parallel corpora, we also construct and incorporate Mango's proprietary high-quality audiovisual subtitle translation dataset for SFT and RL training. This dataset contains over 7 million parallel sentence pairs sourced from Mango TV's exclusive translation resources, covering diverse audiovisual genres including modern dramas, period costume dramas, variety shows, news documentaries and more. It supports 11 key low-resource languages such as Thai, Malay, Indonesian, English and Vietnamese. All content within this dataset undergoes professional human refined translation, which closely aligns with real-world subtitle translation scenarios and complies with standard constraints on subtitle length and rhythm. It delivers critical support for boosting model performance on practical audiovisual subtitle translation tasks. Additionally, we manually annotate more than 800,000 parallel pairs for reinforcement learning fine-tuning.

2.2 Data Processing

The collected data is organized into multiple parallel corpus files spanning English-to-target-language translation directions. All samples are stored in JSON format, which records source texts alongside their corresponding reference translations. Centered on automated translation and automatic evaluation, our data processing pipeline leverages batch translation, asynchronous concurrent processing, and COMET-based quality filtering. Without any manual annotation involvement, we construct high-quality, multilingual aligned parallel corpora through this workflow.

3 Continued Pre-training

At this stage, we adopt the open-source Qwen3.5-9B as the base model to conduct continued pre-training optimized for multilingual audiovisual subtitle translation. The training objective is to strengthen the model's modeling capacity for film dialogues, colloquial expressions, contextual dependency, cross-lingual alignment and low-resource language representations, while preserving the base model's inherent capabilities of general reasoning, instruction comprehension and multilingual knowledge. To boost training efficiency and stability, we employ distributed training, mixed-precision computation and full-parameter training strategies.

3.1 Continued Pre-training Data

To balance general language modeling capability, audiovisual domain adaptability and cross-lingual translation alignment, we adopt a multi-source data mixture strategy in the continued pre-training stage. The corpora cover all 11 target languages, including high-resource languages such as Chinese and English, as well as Thai, Malay, Indonesian, Vietnamese, Japanese, Korean, Arabic, French, Spanish and Russian. The raw data are collected from diverse domains: news articles, film and TV subtitles, variety show & drama dialogues, documentary scripts, web text, encyclopedic knowledge, travel & daily service content, colloquial social media text, and public parallel corpora. This mixed corpus setup brings two clear benefits. First, it broadens the model's coverage of factual knowledge, cultural expressions, colloquial conventions and domain-specific vocabulary across all languages. Second, it prevents catastrophic forgetting of the base model's general linguistic capacity during domain adaptation. We select and clean pre-training corpora from open-source parallel datasets, with a total volume of approximately 200B tokens. The overall data composition is detailed as follows:

- **40% General Chinese Monolingual Corpus**: It covers Chinese news, encyclopedias, web texts, film and television dialogues, variety show colloquial utterances, documentary scripts, and daily service texts. This corpus enhances the model's ability to model Chinese audiovisual texts, colloquial expressions, narrative context, character relationships and cultural backgrounds, while preserving its inherent Chinese comprehension and generation capabilities.
- **10% General Multilingual Corpus**: It contains monolingual raw texts across 11 target languages, including news, encyclopedias, web pages, audiovisual subtitles, travel & daily service content, and informal social utterances. Such data strengthens the model's mastery of vocabulary, syntax, cultural knowledge and conventional expressions in each language, laying a solid foundation for stable target-language generation in cross-lingual translation.
- **50% High-Quality Parallel Corpus**: It consists of open-source parallel corpora, subtitle-aligned parallel data, and proprietary audiovisual translation corpora from Mango TV, forming the core component of continued pre-training. This data enables the model to directly learn cross-lingual mappings, translation patterns, subtitle contextual alignment, and multilingual expression transformation.

3.2 Continued Pre-training Process

Self-Supervised translation learning is adopted in the continued pre-training stage. For all parallel corpora, source-language texts are taken as model input and target-language texts as output; the model is trained with language tags and a unified prompt template.

During continued pre-training, we perform full-parameter training via self-Supervised Fine-Tuning. To guarantee stability and efficiency for large-scale model training, we leverage a distributed training framework to accelerate computation and optimize video memory usage. A conservative learning rate schedule is applied throughout optimization, allowing model parameters to gradually adapt to the data distribution of translation tasks without disrupting the model's inherent linguistic capabilities. A warm-up mechanism at the initial training phase further improves overall training stability. For batch size configuration, gradient accumulation is adopted to enhance the reliability of gradient estimation.

The entire continued pre-training phase ensures sufficient model convergence while balancing general linguistic competence and translation performance, laying a solid foundation for subsequent optimization tailored to audiovisual subtitle scenarios.

4 Post-training

After pre-training, we perform supervised fine-tuning (SFT) followed by reinforcement learning (RL). Specifically, we leverage in-house human-labeled parallel corpus of TV series to conduct supervised fine-tuning on the model, and then carry out RL alignment training with high-quality annotated TV series data. The core objective of this stage is to fully leverage high-quality human-refined drama datasets accumulated by Mango TV, empowering the model with stronger adaptability in complex context comprehension, multi-sentence context modeling, and compliance with subtitle format constraints.

Prompt Template:
你是一个影视剧语言翻译专家，擅长根据影视剧对白并结合语境把中文翻译成其他语言 给定影视剧的一段上下文对白，结合当前影视剧的语境,将指定的对白翻译成阿拉伯语
格式要求 输入输出格式必须完全一致：JSON格式，键为字幕编号，值为翻译内容
- 确保翻译后的条目数量与原始字幕完全一致
- 对专有名词、术语严格按照术语表执行
- 给定上下文，翻译当前文本，不要遗漏任何字幕
输出格式 严格按照以下JSON格式输出，不得有任何额外内容:
```
‘‘‘json ‘\n‘ {{ ‘\n‘ "1": "翻译内容1",‘\n‘ "2": "翻译内容2",‘\n‘ ...
}}‘\n‘ ‘‘‘
```

4.1 Supervised Fine-Tuning

We rely on Mango TV's proprietary translation resources to compile over 7 million parallel corpus pairs. Each sample consists of task instructions, contextual input, and target output. The input segment contains not only the subtitle to be translated but also preceding dialogues as explicit contextual information, enabling the model to fully account for contextual consistency and semantic coherence during generation. The output adheres strictly to structured JSON formatting to meet the requirements of real-world production systems.

We perform full-parameter supervised fine-tuning on the model. A low initial learning rate, paired with a smooth learning rate schedule and warm-up mechanism, is adopted throughout training to prevent excessive disruption to the general translation capabilities acquired in the pre-training phase. After multiple training rounds, the model gradually converges on film and television domain corpora, substantially boosting its accuracy and robustness in real subtitle translation tasks.

4.2 Reinforcement Learning

The SFT phase enables the model to learn the overall distribution of human translations, yet it cannot explicitly distinguish the quality gap between multiple valid translations. It also struggles to directly optimize expressiveness, character tone, and industrial formatting constraints unique to subtitle scenarios. For this reason, we introduce hybrid reinforcement learning alignment after SFT. Instead of relearning fundamental translation capabilities, this phase further aligns the model with human preferences and automatic quality signals atop the SFT base model, producing outputs tailored for real-world audiovisual subtitle delivery.

The first step is human feedback-based preference alignment. We construct pairwise preference datasets from high-quality audiovisual subtitle samples, where each instance contains two candidate translations (chosen and rejected) for a single input prompt. The chosen responses are professionally polished human subtitles, while rejected counterparts are inferior outputs generated by commercial large language models. During annotation and filtering, we prioritize criteria including semantic accuracy, colloquial naturalness, contextual consistency, character tone, emotional expression, cultural adaptation, and subtitle readability. Samples with ambiguous preferences or candidates of comparable quality are discarded. Using this dataset, we perform Direct Preference Optimization (DPO) to bias the model toward generating natural, vivid translations consistent with plot context and character speaking styles.

The second step is quality optimization via automatic rewards. For samples requiring stricter control over semantic faithfulness and delivery formats, we sample multiple candidate translations from the model and build composite reward signals incorporating COMET, BLEU/BLEU-2, subtitle format compliance checks, and repetition penalties. We adopt GRPO-style reward optimization: we conduct relative quality comparison across a set of candidates for the same input, reinforcing high-quality outputs and suppressing low-quality ones according to reward values. Unlike the preference alignment stage, this step emphasizes quantifiable metrics: semantic preservation, phrase consistency, matching JSON and subtitle count limits, suppression of empty/missing translations, and elimination of redundant repetitive generation.

Four categories of automatic reward terms are integrated. First, the COMET reward acts as the primary semantic signal, measuring semantic consistency between source text, reference translations, and model outputs. Second, BLEU/BLEU-2 serves as an auxiliary signal for local phrase and surface-level n-gram matching, stabilizing consistency for terminology, short phrases, and brief subtitle segments. Third, the format compliance reward validates JSON validity, consistent subtitle indexing and quantity, empty outputs, missing translations, misalignment errors, and fill-back compatibility. Fourth, a repetition penalty mitigates redundant and duplicated translations that frequently emerge during post-training. We treat BLEU only as a secondary constraint rather than the dominant reward, preventing the model from overfitting to the surface wording of references and sacrificing the natural phrasing required for audiovisual subtitles.

By chaining DPO and GRPO automatic reward optimization, the post-training pipeline of Mango-MT achieves complementary optimization objectives. DPO focuses on human preference alignment to boost naturalness, character tone, emotional rendering and vividness of translations. Meanwhile, automatic reward optimization strengthens semantic accuracy, formatting compliance, and stability for large-scale industrial production. The final model retains the core translation ability acquired in SFT, while generating more contextually appropriate, production-ready subtitles with natural expressive style tailored to audiovisual scenarios.

5 Translation Evaluation

To comprehensively evaluate the model's multilingual translation performance, we conduct extensive experiments on two benchmarks, namely Mango-SubBench and Flores+, covering a total of 11 languages.

5.1 Mango-SubBench

Subtitle translations are required to simultaneously meet multiple criteria including semantic accuracy, natural wording, stable segmentation, valid timestamps, and traceable quality. For this reason, a single automated metric cannot fully capture the overall quality of subtitle translations. To mitigate this critical limitation, we introduce Mango-SubBench, a dedicated benchmark tailored for audiovisual machine translation tasks. Within this benchmark, we construct a novel evaluation framework alongside a high-quality evaluation dataset. Centered on automated assessment, the evaluation framework enables large-scale, reproducible measurement of the overall performance of diverse models and translation batches. The goal of Mango-SubBench is not merely to measure the similarity between model translations and human reference translations; more importantly, it is to verify whether translated outputs satisfy the practical delivery standards for long-form video content going global.

5.1.1 Evaluation Set

The evaluation set is constructed based on real-world film and television subtitle scenarios, covering 11 languages with a total of 8019 sentence-level samples.

Table 1: Sample distribution of the multilingual subtitle test set

Language	Sample Count
Indonesian	1077
English	762
Vietnamese	657
Malay	489
Thai	457
Korean	699
Japanese	665
Arabic	388
French	1026
Spanish	523
Russian	1026
All	8019

5.1.2 Criteria

Automated evaluation is conducted around two core objectives: semantic quality and subtitle structural compliance. For semantic quality, reference-based COMET[Guerreiro et al. [2024]], semantic similarity, and BLEU-2 metrics [Papineni et al. [2002]] are employed to evaluate translation quality from multiple dimensions including neural assessment, semantic proximity, and surface n-gram matching. In terms of subtitle structure, we inspect the count and order of subtitle segments, empty translations, missing translations, misalignment, timestamp validity, and format compliance to determine whether the translated subtitles can be used for backfilling and formal delivery To enable horizontal comparison of different models and translation batches on the same evaluation set, we derive a weighted composite score for automated evaluation with the formula as follows:

$$S = 0.25 \times E + 0.5 \times C + 0.25 \times B \tag{1}$$

In the formula, C denotes subtitle COMET score, E denotes semantic similarity score, B denotes BLEU-2 score, S refers to the weighted composite overall score.The overall score is computed as a weighted combination of semantic similarity, COMET, and BLEU, with weights of 0.25, 0.50, and 0.25, respectively. COMET is assigned the largest weight because it has been shown to correlate more strongly with human judgments and captures translation adequacy and overall quality by considering the source sentence, machine translation, and reference translation jointly. Semantic similarity is used to complement lexical metrics by measuring sentence-level meaning preservation in the embedding space, while BLEU is retained as a traditional lexical overlap metric to reflect n-gram-level consistency with the reference. This weighting scheme therefore emphasizes semantic adequacy and translation quality while still preserving comparability with conventional machine translation metrics.

- **Semantic similarity**: Built upon Qwen3-Embedding-4B[Zhang et al. [2025]], this metric quantifies the semantic alignment between translated text and source sentences, reflecting the model's capacity to reconstruct and comprehend the original meaning. It is well-suited for flexibly phrased texts such as film and television subtitles.
- **COMET**: We adopt the XCOMET-XL[Guerreiro et al. [2024]] neural network model to evaluate the overall multi-dimensional performance of machine translation by incorporating contextual and semantic information. This metric achieves a high correlation with human subjective assessments, making it suitable for subtitle translation scenarios that demand complex contextual comprehension and cross-sentence reasoning.
- **BLEU-2**: It evaluates the exact n-gram matches between candidate translations and reference texts, focusing on the surface-level accuracy and faithfulness of translations. In quality analysis for multilingual news and subtitles, BLEU-2 is particularly sensitive to short sentences and word order consistency.
- **S(Overall Score)**: Based on the above metrics, we compute a weighted sum of scores across all dimensions to obtain the final composite score, which is used to measure the overall performance of the model in real-world audiovisual subtitle translation scenarios.

5.2 FLORES+

We evaluate the multilingual translation performance of our model on FLORES+[NLLB Team et al. [2024]] which is based on FLORES-200[NLLB Team [2022]]. This dataset was originally released by FAIR researchers at Meta under the name FLORES. The data is now being managed by OLDI, the Open Language Data Initiative. The + has been added to the name to disambiguate between the original datasets and this new actively developed version. The data consists of translations primarily from English into over 200 language varieties. The original English sentences were sampled in equal amounts from Wikinews (an international news source), Wikijunior (a collection of age-appropriate non-fiction books), and Wikivoyage (a travel guide).For each language, the dataset has 997 sentences for the dev split and 1012 sentences for the devtest split. The separate blind test set, originally developed by Meta, is not managed by OLDI.

Following official FLORES evaluation protocols, we adopt BLEU [Papineni et al. [2002]]and COMET[Guerreiro et al. [2024]] metrics to quantify translation quality:

- **BLEU**:SentencePiece-normalized BLEU with a unified 256k multilingual SentencePiece tokenizer, eliminating tokenization bias to enable equitable score comparison across all 200 languages.
- **COMET**: We adopt the XCOMET-XL[Guerreiro et al. [2024]] neural network model to evaluate the overall multi-dimensional performance of machine translation by incorporating contextual and semantic information.

We measure translation performance on FLORES+ using BLEU and COMET on 11 languages following official FLORES+ evaluation rules.

6 Experimental Results

To comprehensively verify the multilingual translation performance of our proposed Mango-MT, we conduct quantitative evaluations on two benchmarks covering distinct application scenarios: the public general multilingual corpus Flores+ and our newly constructed audiovisual subtitle benchmark Mango-SubBench. Table 2 reports automatic translation metrics BLEU and

COMET on Flores+, while Table 3 presents four multi-dimensional evaluation indicators tailored for subtitle tasks, including Semantic Similarity, BLEU-2, COMET and the composite comprehensive score S.

As shown in Table 2, We compare our proposed Mango-MT against three mainstream commercial translation systems, DeepSeek-V4-Pro, Gemini-3-Pro and GPT-5.4, across 11 typologically diverse languages on the Flores+ benchmark using BLEU for lexical n-gram matching and COMET for semantic fidelity evaluation; Our model's average COMET of 0.925 is close to industrial baselines, and it achieves the best semantic score on French and matches the top result on English, with only slight weakness on Arabic. Though its overall BLEU lags Gemini-3-Pro and DeepSeek-V4-Pro, Mango-MT beats GPT-5.4 and attains the highest BLEU on Thai while outperforming two competitors on French. Performance gaps mainly lie in low-resource morphologically complex languages, proving Mango-MT delivers competitive or better semantic translation for most language pairs.

We also give the overall evaluation result shown in Figure 2 , Mango-MT consistently outperforms Gemini-3-Pro, DeepSeek-V4-Pro, and GPT-5.4 across all four evaluation metrics. The improvement is particularly pronounced on BLEU-2 and the overall S score, where Mango-MT achieves relative gains of approximately 29.4% and 9.5% over the strongest baseline, respectively. These results suggest that Mango-MT provides better phrase-level matching, stronger semantic preservation, and more robust overall translation quality. While Gemini-3-Pro performs competitively on COMET, Mango-MT still achieves the highest score, demonstrating the effectiveness of our proposed model across both lexical and neural evaluation metrics.

As shown in Table 3, We evaluate our proposed Mango-MT against three leading commercial translation models (DeepSeek-V4-Pro, Gemini-3-Pro, GPT-5.4) on the Mango-SubBench covering 11 languages via four evaluation dimensions: Semantic Similarity, BLEU-2, COMET and the composite overall score S. Across all 11 tested subtitle languages, Mango-MT outperforms every commercial baseline consistently on all four metrics. In terms of Semantic Similarity, Mango-MT achieves higher values for each language, demonstrating stronger capacity to retain complete contextual and cultural meaning within elliptical, colloquial subtitle sentences. For BLEU-2, a metric tailored for fragmented short subtitle utterances to measure natural bigram collocation, Mango-MT delivers substantial gains over competitors on every language pair, with prominent improvements on Malay, Indonesian, Vietnamese and Spanish that are widely used in cross-border subtitle scenarios. COMET results further validate Mango-MT's superior cross-lingual fluency and authentic spoken style; our model reaches the top COMET score on most languages and eliminates rigid literal translation artifacts seen in commercial systems. When aggregated into the comprehensive metric S that integrates semantic fidelity, phrase matching and readability aligned with industrial subtitle standards, Mango-MT attains a higher S value for every single language, with remarkable advantages on high-frequency subtitle languages including Vietnamese (0.94), Indonesian (0.93), Thai (0.91) and Spanish (0.92). These consistent cross-lingual improvements across all core evaluation dimensions firmly prove that our Mango-MT surpasses mainstream commercial models for film and television subtitle translation tasks.

The two tables reveal clear performance differentiation between general-purpose LLMs and our domain-specialized Mango-MT. On the general formal dataset Flores+, our 9B-parameter Mango-MT achieves competitive results comparable to mainstream ultra-large commercial models. When transferred to the vertical subtitle domain on Mango-SubBench, Mango-MT dominates all baselines across every metric and language. This gap confirms that off-the-shelf large language models lack specialized optimization for colloquial audiovisual text, while our complete training workflow combining multilingual pre-training, subtitle instruction tuning and reinforcement learning enables a lightweight 9B model to achieve state-of-the-art subtitle translation performance. Moreover, Mango-MT balances translation accuracy, naturalness and deployment efficiency, which makes it highly applicable for industrial audiovisual translation scenarios.

Table 2: Multilingual translation evaluation results across 11 languages on Flores+

Models	Metric	Russian	Indonesian	Japanese	French	Thai	English	Spanish	Vietnamese	Arabic	Korean	Malay	Avg
DeepSeek-V4-Pro	BLEU	21.110	27.800	29.680	31.150	10.400	33.760	21.100	32.550	16.120	24.910	23.430	24.730
	COMET	0.945	0.940	0.925	0.921	0.894	0.977	0.945	0.921	0.895	0.914	0.908	0.926
Gemini-3-Pro	BLEU	22.840	28.850	31.640	32.840	9.480	34.090	22.410	32.730	17.300	25.220	24.090	25.590
	COMET	0.948	0.943	0.930	0.924	0.903	0.977	0.948	0.930	0.903	0.919	0.914	0.931
GPT-5.4	BLEU	20.260	24.980	27.710	28.070	8.380	30.990	21.390	30.080	13.240	23.110	21.120	22.670
	COMET	0.946	0.945	0.928	0.925	0.904	0.978	0.950	0.928	0.895	0.920	0.919	0.931
Mango-MT	BLEU	20.370	22.060	27.350	32.980	10.620	30.110	21.640	30.060	10.450	20.440	20.040	22.370
	COMET	0.944	0.932	0.923	0.932	0.905	0.978	0.946	0.925	0.870	0.908	0.911	0.925

Table 3: Multidimensional evaluation results on 11-language subtitle on Mango-SubBench

Models	Metric	Malay	Thai	English	Indonesian	Vietnamese	Russian	French	Japanese	Korean	Spanish	Arabic
DeepSeek-V4-Pro	Semantic Similarity	0.77	0.78	0.84	0.75	0.79	0.82	0.83	0.83	0.81	0.81	0.83
	BLEU-2	0.27	0.35	0.37	0.25	0.39	0.29	0.34	0.34	0.21	0.38	0.30
	COMET	0.90	0.89	0.92	0.93	0.93	0.85	0.83	0.87	0.85	0.88	0.88
	S	0.76	0.80	0.85	0.75	0.85	0.76	0.79	0.80	0.70	0.82	0.78
Gemini-3-Pro	Semantic Similarity	0.79	0.82	0.85	0.77	0.81	0.84	0.85	0.84	0.83	0.80	0.83
	BLEU-2	0.30	0.44	0.39	0.31	0.41	0.33	0.38	0.37	0.24	0.39	0.30
	COMET	0.91	0.91	0.93	0.93	0.93	0.87	0.84	0.87	0.85	0.89	0.88
	S	0.78	0.88	0.86	0.79	0.87	0.79	0.82	0.82	0.73	0.83	0.78
GPT-5.4	Semantic Similarity	0.77	0.80	0.83	0.75	0.80	0.83	0.84	0.83	0.82	0.81	0.82
	BLEU-2	0.27	0.38	0.36	0.26	0.38	0.30	0.35	0.36	0.21	0.36	0.24
	COMET	0.90	0.89	0.91	0.93	0.93	0.85	0.83	0.86	0.84	0.88	0.87
	S	0.75	0.83	0.83	0.76	0.84	0.77	0.79	0.81	0.71	0.81	0.73
Mango-MT	Semantic Similarity	0.82	0.84	0.86	0.84	0.86	0.87	0.88	0.86	0.86	0.87	0.84
	BLEU-2	0.40	0.49	0.44	0.49	0.51	0.46	0.48	0.42	0.33	0.50	0.36
	COMET	0.92	0.91	0.93	0.94	0.94	0.88	0.86	0.89	0.88	0.89	0.87
	S	0.86	0.91	0.90	0.93	0.94	0.89	0.90	0.87	0.81	0.92	0.81

7 Related Work

Prior studies show that small models can achieve comparable or superior task-specific performance against ultra-large models when their capabilities are fully exploited. We overcome the performance constraints of 9B LLMs and develop a human-validated model specialized in film and television subtitle translation. We then survey research on task-tailored LLM refinement, centering our discussion on machine translation.

Domain-specialized large language models: Domain specialization aims to enable models to produce outputs with higher accuracy and more professional quality within target domains. Domain-specialized large language models are trained on custom datasets tailored to specific verticals, including code generation[Zhu et al. [2026];Lyu et al. [2026]], medical science[Wang et al. [2026];Bedi et al. [2025]], mathematical reasoning [Shao et al. [2024],Wang et al. [2024]],

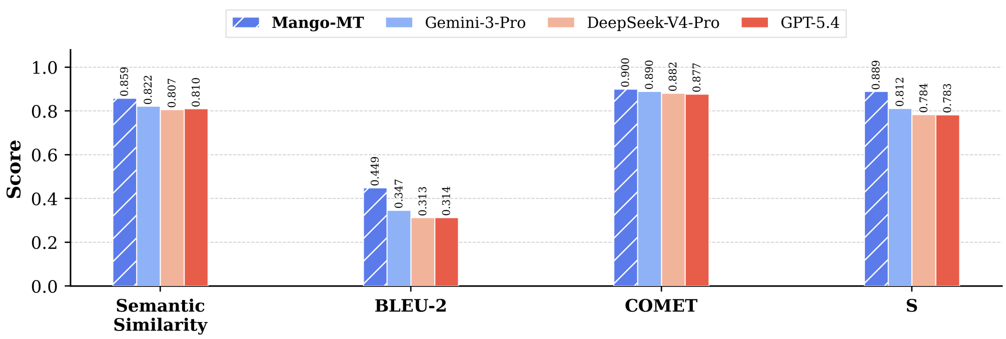

Figure 2: Overall Evaluation Results of 11 Languages on the Mango-SubBench

and finance[Yang et al. [2026]].Two mainstream pipelines are available to develop such models: (1) full pre-training from scratch on a proportionally mixed corpus of general text and domain-specific data; (2) adapting an open-source base model to the target domain via continued pre-training and task-specific supervised fine-tuning, with LoRA as a representative lightweight tuning technique [Hu et al. [2022]]. The second approach drastically cuts training costs, yet the overall model performance is partially constrained by the inherent capabilities of the base model.Accordingly, we opt for full pre-training from scratch to fully unlock the upper bound of multilingual representation during the pre-training phase. Afterward, we conduct supervised fine-tuning to boost the model's professional translation proficiency, followed by reinforcement learning fine-tuning to break the rigid constraints of reference translations and generate more natural, idiomatic target text.

Translation-Specialized LLMs: Numerous studies have strived to match the state-of-the-art multilingual translation performance of GPT-4. TowerInstruct-13B [Alves et al. [2024]] outperforms other open-source counterparts yet still underperforms GPT-4 across automatic evaluation metrics. The ALMA series [Xu et al. [2024]] surpasses NLLB-54B [NLLB Team [2022]] and GPT-3.5, but a clear performance gap relative to GPT-4 remains. Notably, existing literature only conducts automatic evaluations on public machine translation datasets. When human assessments are performed on more diverse, real-world text corpora, however, ultra-large models generally deliver superior performance, especially in challenging translation scenarios [Jiao et al. [2023]]. Our model achieves substantial performance gains and outperforms ultra-large models on automatic metrics. We additionally construct and release a challenging machine translation benchmark tailored for audiovisual subtitles, supporting multi-dimensional evaluation covering expressiveness flexibility, colloquial authenticity, and contextual dependency. Furthermore, this paper presents an efficient training paradigm to fully unlock the model's multilingual translation potential for audiovisual subtitle tasks, and offers a novel research perspective for low-resource language translation.

8 Conclusion

In this paper, we introduce Mango-MT, a professional multilingual machine translation model built exclusively for audiovisual content, which supports translation across 11 languages. Despite only having 9B parameters, Mango-MT delivers translation performance on par with or even superior to state-of-the-art commercial translation systems, as validated by multiple automatic evaluation datasets. We also release the model weights of Mango-MT publicly, aiming to provide an accessible, off-the-shelf tool to facilitate academic research and industrial applications in multilingual subtitle translation. We elaborate on our full training pipeline, including data curation, model pre-training, instruction tuning and reinforcement learning. Furthermore, to evaluate subtitle translation models from diverse perspectives and overcome the limitations of evaluation schemes relying solely on single metrics, we construct a dedicated benchmark named Mango-SubBench, integrated with a comprehensive evaluation framework and high-quality datasets. It assesses subtitles against real-world industrial delivery standards to enable holistic and reliable comprehensive evaluation of translation quality, filling the gap in specialized evaluation resources for multilingual audiovisual subtitle translation.

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